SOLARTRACE

Week 3 Progress Report Hardware Integration & Testing

ESP32 Board Setup & Component Testing

Alek Madut Malual - Lead Developer

Clementina Agiu Gual Koch - UI/UX & Documentation

Abuk Santino Garang Chan - Testing & Logistics

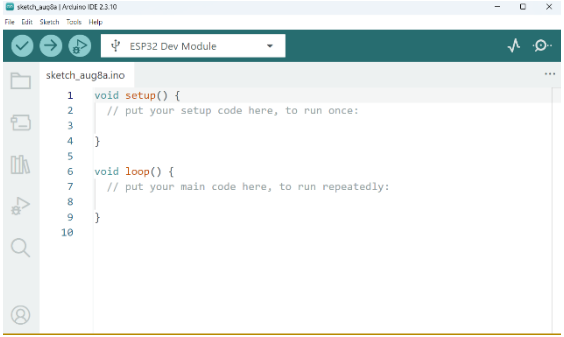
Objectives

- ESP32
- Board Setup
- Arduino IDE
- Configuration
- ESP32 Board
- Addition LED Blink
- Test Component

All Week 3 Objectives Completed Successfully!

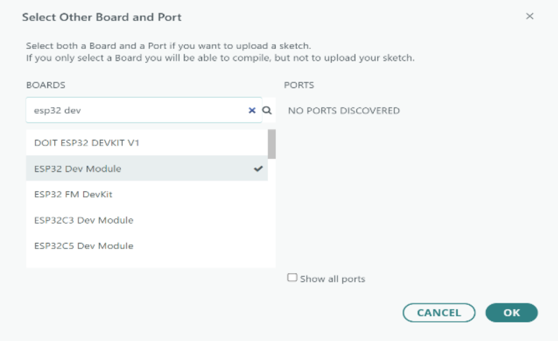
TECHNOLOGY

Step 1: Arduino IDE



Arduino IDE Installed

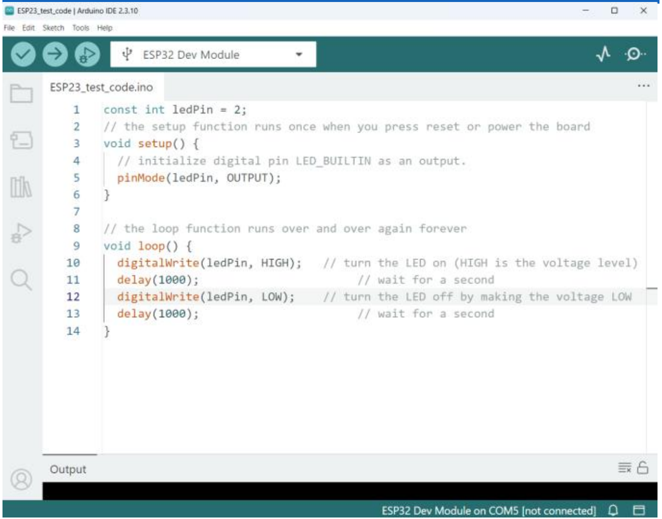
Step 2: ESP32 Board



Board Package Added

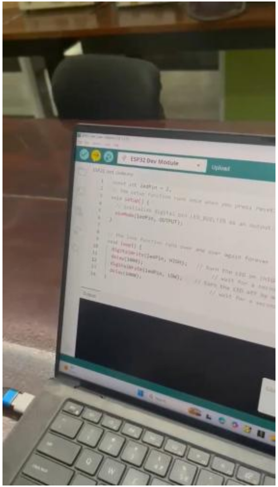
DEMO

Blink Test Code




```
const int ledPin = 2;
// the setup function runs once when you press reset or power the board
void setup() {
  // initialize digital pin LED_BUILTIN as an output.
  pinMode(ledPin, OUTPUT);
}

// the loop function runs over and over again forever
void loop() {
  digitalWrite(ledPin, HIGH);   // turn the LED on (HIGH is the voltage level)
  delay(1000);                       // wait for a second
  digitalWrite(ledPin, LOW);    // turn the LED off by making the voltage LOW
  delay(1000);                       // wait for a second
}
```



Done Test Passed LED blinking at 500ms interval

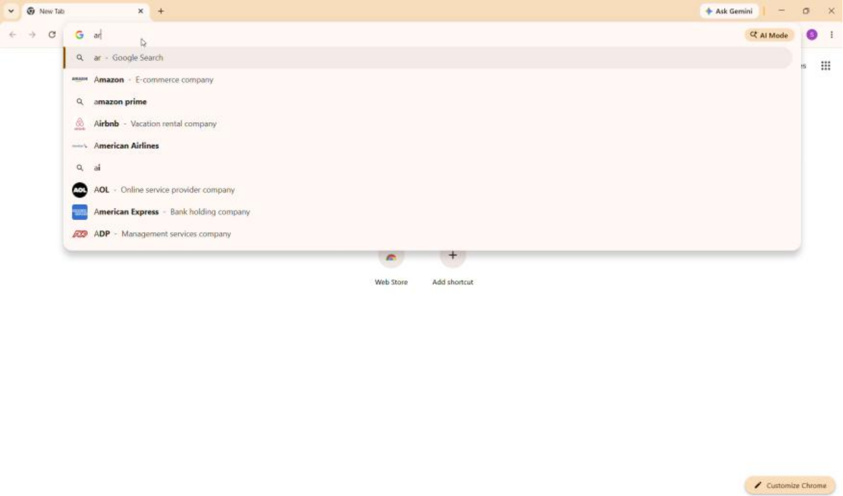
DEMO

Tested Components

- Done ESP32 Development Board
- Done DS3231 RTC Module
- Done MAX6675 Thermocouple
- Done MG995 Servo Motor
- Done MicroSD Card Module
- Done Blynk IoT Connection

DEMO

Video 1: Environment Setup

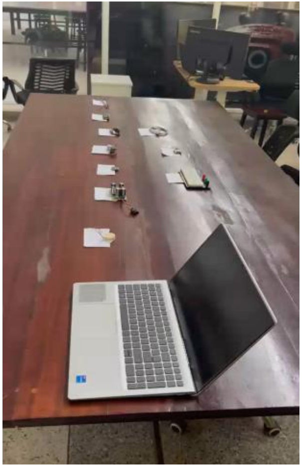


DEMO

Video 2: ESP32 Blink

DEMO

Video 3: All Components



Video files available in project repository

github.com/team-southsudan/solartrace

NEXT STEPS

Week 4 Tasks

Circuit Assembly

- Connect all components
- Verify wiring
- Test power supply

Code Integration

- Combine component codes
- Implement tracking algorithm
- Test all functions

Week 4 Tasks (cont.)

Blynk Dashboard

- Create mobile interface
- Set up notifications
- Test remote monitoring

System Testing

- Full system integration
- Performance testing
- Documentation

Phase I - Week 4 Goal: Complete Prototype Assembly

Integration of all components for SolarTrace tracking system

What We Achieved

- Done Arduino IDE installed
- Done ESP32 board added
- Done LED Blink test passed
- Done All components tested
- Done Documentation complete

Tools & Technologies

- Arduino IDE
- ESP32 Dev Module PlatformIO
- USB Serial Monitor GitHub Repository

Ready for Week 4 - Full Prototype Assembly

SOLARTRACE

Week 3 Progress Report Hardware Integration Complete

Thank You!

Alek Madut Malual Lead Developer	**Clementina Agiu Gual Koch** UI/UX & Documentation	**Abuk Santino Garang Chan** Testing & Logistics

UbuntuNet Alliance Women Hackathon 2026 • August 2026 • Phase I - Week 3

#SolarTrace #CleanEnergy #SouthSudan #UbuntuNetHackathon